

This document outlines the basic steps required to run the LeNet Model on FPGA Project

Required Software:

Quartus Prime Standard Edition 20.1, with ModelSim and Cyclone V device support

Zip file extraction software of your choice

The project is split into four Quartus Prime project files, each contained inside its own folder. Each project implements the full LeNet algorithm with a specific Systolic Array configuration (Row, Output, Input, and Weight stationary). The files will need to be extracted from the zip file. The specific files which run the program are also contained inside these folders, and can be viewed inside of Quartus.

Here is a guide to the folders:

Full_Project → Contains the Weight Stationary Version

Full_Project_IP → Contains the Input Stationary Version

Output_Stationary_Full_project → Contains the Output Stationary Version

Row_Stationary_Full_project → Contains the Row Stationary Version

To launch the project simply open the relevant folder and run the relevant Quartus Project file. This project will be the only file in the folder with the .qpf extension. This can also be done within the Quartus Prime software using File > Open Project, and navigating to the downloaded project.

Several settings were changed to allow for successful compilation, and while these settings should be saved on the .qpf file, they are documented here. In Assignments > Settings > Compiler Setting > Advanced Settings (Synthesis), the following settings have been modified.

1. Allow Register Merging is set to OFF
 - a. This prevents analysis and synthesis from getting stuck at 47%
2. Iteration Limit for constant Verilog loops is increased to 100,000
 - a. This allows the 48000 line Weight Rom memory for FC layer 1 to be correctly initialized without error

Additionally the testbench contains several lines which point to file locations that have likely changed on your laptop. The file “tb_lenet_top” will need to be edited. The file locations in lines 161-2 will need to be updated to your file locations. These can also be found by using ctrl + f and looking for any file locations starting with “C:/”.

At this point the project should compile (Ctrl + L). This will generate the VCD file which had to be deleted, as it is quite large and could not be shared over Canvas and most file sharing platforms.

Key project results can be viewed by running chip planner (Tools > Chip Planner), Power Analyzer (Processing > Power Analyzer Tool), and the RTL Viewer (Tools > Netlist Viewers > RTL Viewer). For the Power analyzer tool, the VCD file in the modelsim folder inside the relevant project file must be assigned using Add Power Input File(s).

The simulated values can be seen by launching the testbench in ModelSim Tools > Run Simulation Tool > RTL Simulation.

In each project the SystemVerilog Files that make up the model itself are contained.

Fc_top.sv, fc_layer.sv, and fc_serial.sv define the fully connected layers

Pooling_top.sv, and pooling_layer.sv define the pooling layers

Systolic_top.sv, PE.sv, and PE_array.sv define the convolution layers

Model_top.sv defines the structure as a whole

And other files define memories throughout the project